

8 Puzzles A* (A Star) using Heuristic Search

```
import heapq

GOAL = (1, 2, 3, 8, 0, 4, 7, 6, 5)

def heuristic(s):
    return sum(a != b and a != 0 for a, b in zip(s, GOAL))

def neighbors(s):
    z = s.index(0)
    r, c = divmod(z, 3)

    for dr, dc in [(-1, 0), (1, 0), (0, -1), (0, 1)]:
        nr, nc = r + dr, c + dc

        if 0 <= nr < 3 and 0 <= nc < 3:
            n = list(s)
            i = nr * 3 + nc
            n[z], n[i] = n[i], n[z]
            yield tuple(n)

def inversions(s):
    a = [x for x in s if x]
    return sum(a[i] > a[j]
                for i in range(len(a))
                for j in range(i + 1, len(a)))

def solvable(s):
    return inversions(s) % 2 == inversions(GOAL) % 2

def astar(start):
    pq = [(heuristic(start), 0, start, [start])]
    visited = {start: 0}

    while pq:
        f, g, state, path = heapq.heappop(pq)
```

```

        if state == GOAL:
            return path
    for n in neighbors(state):
        ng = g + 1
        if n not in visited or ng < visited[n]:
            visited[n] = ng
            heapq.heappush(
                pq, (ng + heuristic(n), ng, n, path + [n])
            )

def print_board(s):
    for i in range(0, 9, 3):
        print(" ".join(str(x) if x else " " for x in s[i:i + 3]))
    print("-" * 9)

while True:
    try:
        s = tuple(map(int, input("Enter 9 numbers: ").split()))

        if len(s) != 9 or sorted(s) != list(range(9)):
            print("Enter numbers 0 to 8 exactly once.\n")
            continue
        break
    except ValueError:
        print("Enter integers only.\n")

print("\nInitial State:")
print_board(s)

print("Goal State:")
print_board(GOAL)

if not solvable(s):
    print("This puzzle configuration is UNSOLVABLE.")
else:
    print("Solvability check passed! Solving using A* Search...\n")
    solution = astar(s)

    if solution:
        print(f"Solution found in {len(solution) - 1} steps!\n")

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        for step, state in enumerate(solution):
            print(f"Step {step}:")
            print_board(state)
    else:
        print("No solution found.")
```

```
PS C:\Users\Student\Desktop\1WA24CS203> & C:\Users\Student\AppData\Local\Python\
les_Astar_Heuristic.py
```

```
Enter 9 numbers: 2 8 3 1 6 4 7 0 5
```

```
Initial State:
```

```
2 8 3
```

```
1 6 4
```

```
7 5
```

```
-----
```

```
Goal State:
```

```
1 2 3
```

```
8 4
```

```
7 6 5
```

```
-----
```

```
Solvability check passed! Solving using A* Search...
```

```
Solution found in 5 steps!
```

```
Step 0:
```

```
2 8 3
```

```
1 6 4
```

```
7 5
```

```
-----
```

```
Step 1:
```

```
2 8 3
```

```
1 4
```

```
7 6 5
```

```
-----
```

```
Step 2:
```

```
2 3
```

```
1 8 4
```

```
7 6 5
```

```
-----
```

```
Step 3:
```

```
2 3
```

```
1 8 4
```

```
7 6 5
```

```
-----
```

```
Step 4:
```

```
1 2 3
```

```
8 4
```

```
7 6 5
```

```
-----
```

```
Step 5:
```

```
1 2 3
```

```
8 4
```

```
7 6 5
```

```
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```